

GEOOptimize

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Proposal for:	Proposal Date:	Proposal Number:
SMS Engineering 770 Bradford Street Winnipeg, MB R3H 0N3 ATTN: Jean-Louis Gratton, P. Eng.	2025-10-07	QU-200
		Reference
		CA MB SMS Winnipeg Schools

Understanding of project

The Manitoba Government proposes to build three new schools, to be constructed in Winnipeg, Manitoba. Each school is expected to be a two-story building with a footprint of approximately 5,500 square meters. Each new school shall demonstrate compliance with Manitoba's Green Building Program, achieve LEED Silver certification, and explore geothermal systems as a source of renewable energy. Two of those schools will feature a closed-loop horizontally bored ground heat exchanger (GHX). They are for:

1. The Pembina Trails School Division on Skyline Drive in Winnipeg, MB, and,
2. Seven Oaks School Division on Meadowland Drive in Winnipeg, MB.

Ground source heat pump (GSHP) systems for each school will be designed by SMS Engineering. *This proposal discusses the opportunity for GEOOptimize to provide design assistance to SMS who will take responsibility for the design at the time of tender. GEOOptimize will provide technical support during construction.*

Scope of work

Design Phase

For each school, GEOOptimize will:

- Assist in development of a Basis of Design and mechanical system schematic.
- Review geotechnical data and available TC Test information for the thermal properties of the soil at each site.
- Create a GHX model using Ground Loop Design (GLD 2016) modelling software that will be used in determining the configuration, size, and flow characteristics of the proposed GHX.
 - The GHX model will be sized using the 8,760-hour energy model created by SMS.
- Provide design details for final GHX, including sketches and written Specifications, to SMS. SMS to create and seal drawings Issued for Construction. Design details to include:
 - GHX circuit design and layout,
 - GHX runout and header design, including reducing tee schedule,
 - GHX manifold design,
 - Description and scope of work including mark of demarcation for GHX sub-contractor,

- Workmanship and site restoration requirements,
- Products and Materials specification,
- Heat transfer fluid specification,
- Installation and coordination requirements,
- Documentation and submission requirements, including submittals, QA/QC, commissioning, including flushing and purging, and warranty.
- The design will include the GHX up to and including the manifold in the building. GEOptimize will provide the flow rate and pressure drop of the GHX so that SMS can select an appropriate pump.

Tendering and Construction Phase

- Support the bidding process by answering questions and helping to review the qualifications of geothermal sub-contractors,
- Attend a pre-construction site meeting with selected GHX Contractor and project stakeholders (not including travel expenses if applicable).
- Support the construction process with respect to RFI's and PCN's,
- Review site photos by SMS or Contractor.
- Attend up to two (2) site inspections during construction, including inspection reports and photographs (not including travel expenses if applicable).

Methodology

The size, cost, and performance of a GHX is sensitive to the peak heat transfer to and from the ground. It is also sensitive to the total heat rejection to the ground when the building is being cooled compared to the amount of heat extracted from the ground to heat the building...the annual energy balance.

Accurate hourly energy modelling of the building(s) connected to a ground heat exchanger (GHX) is critical to the development of a GSHP system that is cost-effective to build, performs efficiently and predictably for the life of the building and is easily maintained. Energy loads are determined by simulating building performance based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data for the location.

Working closely with the architectural and mechanical design team early in the design process provides the opportunity to adjust the energy loads to and from the GHX. Using the hourly energy model (by SMS) to determine the impact of changes to glass specifications, lighting and lighting controls, ventilation air strategy, domestic hot water production, distribution system design, etc. can result in both a reduction in the size and cost of the GHX and an improvement in the efficiency and performance of the system. GEOptimize will review the energy model created by SMS and work closely with their design team to ensure the loads are optimal for GHX design.

Energy efficiency measures resulting in improvement in GHX performance and a reduction in capital cost are reviewed with the design team and owners. Preliminary GHX modelling based on approved changes in the energy model is developed based on site constraints and measured geological conditions to ensure the required land area is available and it can be built cost-effectively with the contractor resources available in the area.

With team agreement on final energy model iterations, energy consumption the technical and economic viability of installing a GSHP system for the project is completed based on:

- Iterative hourly energy modelling developed in collaboration with owner and design team
- GHX design based on review and testing of geological information on site and available land area for construction of GHX for building(s)
- Energy consumption and cost of GSHP system in comparison to the baseline system, based on current utility rate structure
- CO2 emissions of GSHP system in comparison to baseline system

A conductivity test for a horizontal GHX is conducted by collecting soil samples from across the site and at the depth of the proposed GHX with an excavator or with core samples. Thermal conductivity and diffusivity are measured with a Tempos Thermal Conductivity meter. The final GHX design is adjusted based on the measured soil properties.

NOTE: GEOOptimize is not capable of conducting a vertical TC test. A new school in Manitoba is likely to have a playfield and geological conditions suitable for a Horizontally configured GHX. If it is found that a horizontal GHX is not suitable for a particular site GEOOptimize can assist in tendering a vertical TC test instead.

If the feasibility assessment for construction of a GSHP system is acceptable to the client, GEOOptimize will

- Prepare design drawings and specifications for authority approval with the client and ready for SMS to Seal for inclusion with Tender documents.
- Review bid documents and provide construction administration, and assist testing and commissioning of the system with a develop a quality assurance / quality control program with the design team
- Coordinate with SMS as needed on the detailed design of the heat pump system in the facility.

Key Personnel for Project:

Ed Lohrenz

Ed has worked in the GeoExchange™ industry since 1982 as an installation contractor, equipment supplier, manufacturer, and consultant. He is an accredited trainer for IGSHPA for both the Accredited Installer Course and the Certified GeoExchange Designer (CGD) commercial design course. He was vice-chair of the Canadian Standards Association C-448 Standard for the Design and Installation of Earth Energy Systems (2002 edition) and Technical Committee Member ANSI/CSA C448 Standard (2016). He has worked with building owners, mechanical contractors, developers, architects, and engineering consultants across North America and internationally.

Ian Lohrenz

Ian has made a career in the Heating, Ventilating and Air Conditioning (HVAC) industry since 2008. He is a graduate of the University of Manitoba in 2008 with a Bachelor of Science in Mechanical Engineering and has completed specialized training courses in commissioning (NRCan's Advanced Course on Building Recommissioning), energy modelling (Trane Trace 700 and Trane Trace 3D+ computer design simulation courses in La Crosse, WI,) and ground source heat pump system design (Canadian GeoExchange™ Coalition Design Course and Commercial System Design). Capitalizing on his experience in building performance and the interaction between architecture and building systems Ian has developed the capability to optimize the cost and long-term performance of ground sourced heat pump systems through energy modelling, energy management, and commissioning.

Stephen Hak

Stephen has worked in the GeoExchange™ industry since 2020 as system designer consultant. Biosystems Engineering graduate with a specialization in Environmental Engineering. Currently in part time masters program focusing on a thesis to determine the performance of vertical boreholes integrating into structural piles to avoid the extra cost of drilling. Experience in energy modelling, ground heat exchanger modelling, energy systems modelling, consulting, quality control, material testing, construction, supervision, and general contracting. Experience working with mechanical contractors, developers, architects, and engineering consultants across North America and internationally.

Deliverables

Design Phase

- PDF Reports showing the details of the modelled GHX designs,
- Sketches of proposed GHX designs, and written specifications summarizing the GHX design will be prepared for presentation to SMS. SMS will create detailed drawings for Issued for Construction in CAD, seal the designs and provide Contract Administration with support from GEOptimize.

Tendering and Construction Phase

- Pre-Construction Site meeting agenda and minutes.
- Written site inspection report.
- Comments regarding Shop Drawings and RFI's etc. via email.

Required information:

The following information is required as it becomes available to accurately model the performance of the GHX:

- Hourly energy model(s) of the building provided in a .csv file that excludes the electrical energy needed to power the heat pump compressors.
- Site plan clearly defining the land area available for construction of the proposed GHX including site services to be installed.
- Building drawings, specifications, and any available information regarding occupancy and assumed use schedules.
- Geotechnical reports.
- Site plan in both AutoCAD and PDF formats.

Fees

Description: (Base Contract)	School 1	School 2
Design Phase: AutoCAD drawings and written specifications	\$ 19,900	\$ 19,900
Construction Phase: Tender & Construction support, including One (1) pre-construction site meeting and up to two (2) site inspections.	\$ 6,400	\$ 6,400
Sub-Total <i>not including applicable taxes</i>	\$26,300	\$ 26,300
Design Total <i>not including applicable taxes</i>	\$ 52,600	

Total value of all services..... \$ 52,600
Plus taxes,

Terms:

- Additional work out of scope will be charged at \$225/hr plus expenses.
- Additional site meetings or inspections, if required, will be billed at time plus expenses.
- Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

Authorized signature

Terms & Conditions

General terms and conditions for Professional Services in Canada

1. Applicability

- a. These terms and conditions of sale (these "Terms") are the only terms that govern the sale by **GEOptimize, Inc.** ("**Service Provider**") of the services ("**Services**"), as more particularly described on the Service Provider's Proposal to which these Terms are attached (the "Proposal") to the client named on the Proposal ("**Client**"). Service Provider and Client are sometimes referred to in these Terms as a "**Party**" or the "**Parties**."
- b. The accompanying Proposal and these Terms (collectively, this "**Agreement**") comprise the entire agreement between the parties and supersede all prior or contemporaneous understandings, agreements, negotiations, representations and warranties, and communications, both written and oral. These Terms prevail over any of Client's general terms and conditions of purchase regardless of whether or when Client has submitted its purchase order or such terms. Fulfillment of Client's order does not constitute acceptance of any of Client's terms and conditions and does not serve to modify or amend these Terms. Notwithstanding anything herein to the contrary, if a written contract signed by both parties is in existence covering the sale of the Services covered hereby, the terms and conditions of said contract shall prevail to the extent they are inconsistent with these Terms.

2. Performance of Services

- a. *Services.* In accordance with this Agreement and applicable law, Service Provider shall provide to Client the Services set forth in the Proposal.
- b. *Change Orders; Scope Changes.* Any additions to or changes to the scope of Services must be agreed by the parties in a written change order ("Change Order") or other writing. The Parties will equitably adjust pricing and schedule and enter into a Change Order reflecting such adjustments if (i) a scope change is necessary due to emergency, safety, or other reasonable necessity, or (ii) if specific periods of time for rendering Services set forth in the Proposal are exceeded through no fault of Service Provider.
- c. *Client's Obligations.* In connection with the Services, Client shall reasonably cooperate with Service Provider and provide Service Provider with information and access to Client's premises and facilities, as may reasonably be requested by Service Provider.

3. *Delays.* If Service Provider's performance of its obligations under this Agreement is prevented or delayed by any act or omission of Client or its agents, subcontractors, consultants, or employees ("**Client Caused Delay**"), Service Provider shall not be deemed in breach of its obligations under this Agreement or otherwise liable for any costs, charges, or losses sustained or incurred by Client, in each case, to the extent arising directly or indirectly from such prevention or delay. In addition, Service Provider shall not be deemed in breach of its obligations under this Agreement for any delays to the extent resulting from circumstances or events outside of Service Provider's control.

4. Fees and Expenses

- a. *In General.* In consideration of the provision of the Services by the Service Provider and the rights granted to Client under this Agreement, Client shall pay the fees ("**Fees**") as follows:
 - i. *Time and Materials.* For any Services provided on a time and materials basis, as specified in the Proposal, Client shall pay Fees calculated in accordance with Service Provider's daily/hourly fee rates as set forth in the Proposal.
 - ii. *Fixed Price.* Where Services are provided for a fixed price, as specified in the Proposal, the total Fees shall be the amount set out in the Proposal.
- b. *Travel Expense Reimbursement.* Client shall reimburse Service Provider for any travel necessary to perform the Services as specified in the Proposal or as otherwise approved by Client in writing.
- c. *Taxes.* All Fees are exclusive of all sales, use, value added and excise taxes, and any other similar taxes, duties, and charges of any kind imposed by any governmental authority on any amounts payable by Client. Client shall be responsible for all such charges, costs, and taxes. To the extent Service Provider is required to pay any such sales, use, excise, value added or other taxes or duties or charges, Client shall reimburse Service Provider for such payments. Notwithstanding the foregoing,

Client shall not be responsible for any taxes imposed on, or with respect to, Service Provider's income, revenues, gross receipts, personal or real property, or other assets.

- d. **Collection Expenses.** If Service Provider incurs any costs, expenses, or fees, including reasonable legal fees and professional collection services fees, in connection with the collection or payment of any amounts due it under this Agreement, Customer agrees to reimburse Service Provider for all such costs, expenses and fees.

- 5. **Payment Terms.** Unless otherwise specified in the Proposal, Client shall pay Fees within 30 days of an invoice from Service Provider. Invoices that remain unpaid for more than 15 days after the payment due date may, at Service Provider's discretion, bear interest at the rate of one percent (1%) per month or the maximum rate permitted by law, whichever is less, calculated daily and compounded monthly. Client shall also reimburse Service Provider for all costs it incurs in collecting any late payments, including, without limitation, reasonable attorneys' fees and court costs. In addition to all other remedies available under the Agreement or at law, Service Provider shall be entitled to suspend the performance of any Services if Client fails to pay any amounts when due hereunder and such failure continues for 30 day following written notice thereof, and such suspension of Services shall not be considered a breach or default of any of Service Provider's obligations under this Agreement.

- 6. **Limited Warranty**

- a. Service Provider warrants that it shall provide the Services (a) in accordance with the terms of this Agreement; (b) using personnel of industry standard skill and experience who are qualified and licensed, as required; and (c) in a timely, workmanlike, and professional manner in accordance with generally recognized industry standards for similar services.
- b. Except for the warranties set forth in section 6(a), Service Provider makes no warranty whatsoever with respect to goods or services, including any (a) warranty of merchantability; (b) warranty of fitness for a particular purpose; or (c) warranty against infringement of intellectual property rights of a third party; whether express or implied by law. Service Provider shall not be liable for any services or work product or deliverables provided by third-party vendors identified or referred to the Client by the Service Provider during the term of this Agreement, pursuant to any Proposal or otherwise, except if such third-party services are provided under written agreement with Service Provider.
- c. Service Provider shall not be liable for a breach of the warranties set forth in Section 6(a) unless (i) Client gives written notice of the defective Services, reasonably described, to Service Provider within 90 days after the completion of the Services and (ii) Service Provider reasonably verifies Client's claim that the Services are defective.
- d. With respect to any Services subject to a claim under the warranty set forth in Section 6(a), Service Provider shall, in its sole discretion, (i) repair or re-perform the applicable Services, without additional charge or (ii) credit or refund the price of such Services attributable to the applicable Services.
- e. The remedies set forth in section 6(d) shall be the Client's sole and exclusive remedy and Service Provider's entire liability for any breach of the limited warranty set forth in section 6(a).

- 7. **Limitation of Liability**

- a. In no event shall Service Provider be liable to Client or any third party for any consequential, indirect, incidental, special, exemplary, or punitive damages whether arising out of breach of contract, tort (including negligence), strict liability or otherwise, regardless of whether such damages were foreseeable and whether or not Service Provider has been advised of the possibility of such damages, and notwithstanding the failure of any agreed or other remedy of its essential purpose.
- b. In no event shall Service Provider's aggregate liability arising out of or related to this agreement, whether arising out of or related to breach of contract, tort (including negligence), strict liability or otherwise, exceed the total of the amounts paid to Service Provider for the services provided under this Agreement.

8. **Intellectual Property:** All intellectual property rights, including copyrights, patents, patent disclosures and inventions (whether patentable or not), trademarks, service marks, trade secrets, know-how and other confidential information, trade dress, trade names, logos, corporate names and domain names, together with all of the goodwill associated therewith, derivative works and all other rights (collectively, "**Intellectual Property Rights**") in and to all documents, work product and other materials that are delivered to Client under this Agreement or prepared by or on behalf of the Service Provider in the course of performing the Services, including any items identified as such in the Proposal (collectively, the "**Deliverables**"), except for any Confidential Information of Client or Client materials, shall be the sole and exclusive property of the Service Provider. Subject to Client's payment of the Fees, Service Provider hereby grants Client, a license to use all Intellectual Property Rights in the Deliverables free of additional charge and on a non-exclusive, worldwide, non-transferable, non-sublicensable, fully paid-up, royalty-free and perpetual basis to the extent necessary to enable Client to make reasonable use of the Deliverables and the Services. The Client shall indemnify, defend and hold Service Provider harmless from and against all claims, actions or suits arising out of or related to the Client's unauthorized use of the Deliverables or infringement upon or misappropriation of Service Provider's Intellectual Property Rights.
9. **Term:** Termination and Survival
- a. This Agreement shall commence as of the Effective Date and shall continue thereafter until the completion of the Services (the "Term") unless sooner terminated pursuant to paragraphs (b) or (c) of this Section 9.
 - b. Either Party may terminate this Agreement, effective upon written notice to the other Party (the "Defaulting Party") if the Defaulting Party:
 - i. Breaches this Agreement (other than a payment default governed by paragraph (c) of this Section), and the Defaulting Party does not cure such breach within 30 days after receipt of written notice of such breach, or such breach is incapable of cure; or
 - ii. Experiences a bankruptcy, insolvency, or financial insecurity event in which it: (i) becomes or is declared insolvent or bankrupt; (ii) is the subject of a voluntary or involuntary bankruptcy or other proceeding related to its liquidation or solvency, which proceeding is not dismissed within 10 calendar days after its filing; (iii) ceases to do business in the normal course; or (iv) makes an assignment for the benefit of creditors.
 - c. Client shall pay Service Provider for all Services rendered prior to the effective date of termination.
10. **Waiver.** No waiver of any term or right in this Agreement shall be effective unless in writing, signed by an authorized representative of the waiving Party. The failure of either Party to enforce any provision of this Agreement shall not be construed as a waiver or modification of such provision, or impairment of its right to enforce such provision or any other provision of this Agreement thereafter.
11. **Confidential Information.** From time to time during the Term of this Agreement, either Party (as the "**Disclosing Party**") may disclose or make available to the other Party (as the "**Receiving Party**"), non-public, proprietary, and confidential information of Disclosing Party ("**Confidential Information**"); provided, however, that Confidential Information does not include any information that: (a) is or becomes generally available to the public other than as a result of Receiving Party's breach of this **Error! Bookmark not defined.**; (b) is or becomes available to the Receiving Party on a non-confidential basis from a third-party source, provided that such third party is not and was not prohibited from disclosing such Confidential Information; (c) was in Receiving Party's possession prior to Disclosing Party's disclosure hereunder; or (d) was or is independently developed by Receiving Party without using any Confidential Information. The Receiving Party shall: (x) protect and safeguard the confidentiality of the Disclosing Party's Confidential Information with at least the same degree of care as the Receiving Party would protect its own Confidential Information, but in no event with less than a commercially reasonable degree of care; (y) not use the Disclosing Party's Confidential Information, or permit it to be accessed or used, for any purpose other than to exercise its rights or perform its obligations under this Agreement; and (z) not disclose any such Confidential Information to any person or entity, except to the affiliated entities, directors, officers, members, managers, employees, contractors,

agents, advisors, consultants, financing parties, tax equity investors, or lenders of Receiving Party or its affiliated entities (“**Representatives**”) who need to know the Confidential Information to assist the Receiving Party, or act on its behalf, to exercise its rights or perform its obligations under this Agreement. The Receiving Party agrees that any of its Representatives to whom Confidential Information is disclosed will be informed of the confidential nature of the information and the Receiving Party’s confidentiality obligations under this Agreement, and will be subject to a contractual or other obligation to protect the Confidential Information upon terms consistent with the requirements of this Section. Each Party agrees to be responsible for any act or omission by its Representatives to whom it has disclosed Confidential Information under this Agreement if such act or omission, if done by the Receiving Party, would constitute a breach of this Agreement. If the Receiving Party is required by applicable law or legal process to disclose any Confidential Information, it shall, prior to making such disclosure, use commercially reasonable efforts to notify Disclosing Party of such requirements to afford Disclosing Party the opportunity to seek, at Disclosing Party’s sole cost and expense, a protective order or other remedy. The Parties agree that they each shall be entitled to, in addition to other available remedies, to seek specific performance and further injunctive relief to prevent or restrain breaches or threatened breaches of this Section. The confidentiality obligations set forth in this Agreement shall continue for a period of two (2) years after the completion of the Services. The Receiving Party shall not decompile, disassemble, or reverse engineer all or any part of the Confidential Information.

12. Subcontracting. Service Provider shall be responsible for Services performed by any subcontractor. Service Provider may subcontract any part of its obligations under the Agreement.
13. Assignment. Neither Party shall assign any of its rights or delegate any of its obligations under this Agreement without the prior written consent of the other Party (not to be unreasonably withheld), provided however, Service Provider shall have the right to freely assign this Agreement to an affiliated entity or to a successor in interest in connection with a merger, acquisition, or sale of all or substantially all of the assets of Service Provider, or other similar strategic transaction affecting Service Provider. Upon any such assignment, Service Provider will provide prompt written notice to Client. Any purported assignment or delegation in violation of this Section is null and void.
14. Relationship of the Parties. With respect to any Services provided under this Agreement, the Service Provider is an independent contractor and not an employee of the Client. Nothing contained in this Agreement shall be construed as creating any agency, partnership, joint venture, or other form of joint enterprise, employment, or fiduciary relationship between the Parties, and neither Party shall have authority to contract for or bind the other party in any manner whatsoever.
15. No Third-Party Beneficiaries. This Agreement is for the sole benefit of the Parties hereto and their respective successors and permitted assigns and nothing herein, express or implied, is intended to or shall confer upon any other person or entity any legal or equitable right, benefit, or remedy of any nature whatsoever under or by reason of the Agreement.
16. Governing Law and Jurisdiction. This Agreement shall be governed by and construed in accordance with the laws of Manitoba and any court of competent jurisdiction in Manitoba shall have jurisdiction to adjudicate any matter arising out of this Agreement. Both Parties hereby submit to the exclusive jurisdiction and venue of any such court.
17. Severability. If any term or provision of this Agreement is invalid, illegal, or unenforceable in any jurisdiction, such invalidity, illegality, or unenforceability shall not affect any other term or provision of this Agreement or invalidate or render unenforceable such term or provision in any other jurisdiction.
18. Survival. Each term and provision of this Agreement that should by its sense and context survive any termination or expiration of this Agreement, shall so survive regardless of the cause and even if resulting from the material breach of either Party to this Agreement.
19. Amendment and Modification. This Agreement, and any exhibits attached hereto, is the entire agreement between the Parties with respect to the subject matter hereof and supersedes any prior agreement or communications between the Parties, whether written, oral, electronic or otherwise. These Terms may only

be amended or modified in a writing stating specifically that it amends these Terms and is signed by an authorized representative of each party.

20. Rights Cumulative. The rights and remedies of the Parties herein provided shall be cumulative and not exclusive of any rights or remedies provided by law or equity.
21. Acceptance of Terms. Client shall be deemed to have accepted these Terms if (a) Client signs and returns the Proposal to Service Provider, or (b) Client accepts the performance of Services from Service Provider after Client has had a reasonable opportunity to review the Proposal and these Terms, whichever occurs first.

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